

## 1. Online Convex Optimization

## 1. Framework

**Decision space:** Closed convex set  $\mathcal{K} \subseteq \mathbb{R}^d$

**Protocol** (for  $t = 1, \dots, T$ ):

1. Learner chooses  $x_t \in \mathcal{K}$
2. Environment reveals convex loss  $f_t : \mathcal{K} \rightarrow \mathbb{R}$
3. Learner incurs loss  $f_t(x_t)$

**Regret:**

$$R_T = \sum_{t=1}^T f_t(x_t) - \min_{x \in \mathcal{K}} \sum_{t=1}^T f_t(x)$$

**Goal:** Achieve  $O(\sqrt{T})$  regret.

## 2. Examples

**Portfolio optimization:**

- $\mathcal{K} = \{x \in \mathbb{R}^n : x_i \geq 0, \sum_i x_i = 1\}$  (simplex)
- $f_t(x) = -\langle r_t, x \rangle$  (negative return)

**Online regression:**

- $\mathcal{K} = \{x \in \mathbb{R}^d : \|x\| \leq B\}$  (ball)
- $f_t(x) = (y_t - \langle x, \varphi(x_t) \rangle)^2$  (squared loss)

**Parameter tuning:**

- $\mathcal{K} = [0, 1]^d$  (box)
- $f_t(x)$  = validation error with hyperparameters  $x$

## 2. Online Gradient Descent (OGD)

## 1. Algorithm

**Parameters:** Step size  $\eta_t > 0$

**Initialize:**  $x_1 \in \mathcal{K}$  arbitrarily

**For**  $t = 1, \dots, T$ :

1. Play  $x_t$
2. Observe  $f_t$  and compute  $\nabla f_t(x_t)$
3. Update:

$$\tilde{x}_{t+1} = x_t - \eta_t \nabla f_t(x_t)$$

$$x_{t+1} = \Pi_{\mathcal{K}}(\tilde{x}_{t+1})$$

where  $\Pi_{\mathcal{K}}$  is Euclidean projection onto  $\mathcal{K}$ .

**Typical choice:**  $\eta_t = \frac{D}{G\sqrt{t}}$  where  $D = \text{diam}(\mathcal{K})$ ,  $G = \sup_t \|\nabla f_t(x_t)\|$ .

## 2. Regret Bound

**Theorem:** If  $\|x - y\| \leq D$  for all  $x, y \in \mathcal{K}$  and  $\|\nabla f_t(x)\| \leq G$ , then OGD with  $\eta = \frac{D}{G\sqrt{T}}$  achieves:

$$R_T \leq \frac{3}{2} DG\sqrt{T}$$

**Key properties:**

- $O(\sqrt{T})$  regret (optimal)
- No convexity of  $\mathcal{K}$  needed beyond projection
- Works for any convex losses  $f_t$

## 3. Proof Sketch

**Key inequality:** For convex  $f$ ,

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle$$

**Step 1:** Bound one-step regret using projection property.

**Step 2:** Telescoping sum over distance  $\|x_{t+1} - x^*\|^2$ .

**Step 3:** Optimize step size  $\eta$  to balance terms.

**Result:** Regret scales as  $\sqrt{T}$ .

## 3. Projection onto Convex Sets

## 1. Definition

**Euclidean projection:**

$$\Pi_{\mathcal{K}}(y) = \arg \min_{x \in \mathcal{K}} \|x - y\|_2$$

This is the closest point in  $\mathcal{K}$  to  $y$ .

**Properties:**

1. Unique (if  $\mathcal{K}$  is closed and convex)
2. Non-expansive:  $\|\Pi_{\mathcal{K}}(y) - \Pi_{\mathcal{K}}(z)\| \leq \|y - z\|$
3. Satisfies:  $\langle y - \Pi_{\mathcal{K}}(y), x - \Pi_{\mathcal{K}}(y) \rangle \leq 0$  for all  $x \in \mathcal{K}$

## 2. Common Projections

**Ball:**  $\mathcal{K} = \{x : \|x\| \leq R\}$

$$\Pi_{\mathcal{K}}(y) = \begin{cases} y & \text{if } \|y\| \leq R \\ \frac{Ry}{\|y\|} & \text{otherwise} \end{cases}$$

**Simplex:**  $\mathcal{K} = \{x : x_i \geq 0, \sum_i x_i = 1\}$

- Use sorting algorithm in  $O(d \log d)$

**Box:**  $\mathcal{K} = [a, b]^d$

$$\Pi_{\mathcal{K}}(y)_i = \max(a, \min(b, y_i))$$

**Halfspace:**  $\mathcal{K} = \{x : \langle a, x \rangle \leq b\}$

$$\Pi_{\mathcal{K}}(y) = y - \max\left(0, \frac{\langle a, y \rangle - b}{\|a\|^2}\right) a$$

## 4. Follow-The-Regularized-Leader (FTRL)

## 1. Algorithm

**Idea:** Minimize cumulative loss plus regularization.

**At round  $t$ , play:**

$$x_t = \arg \min_{x \in \mathcal{K}} \left[ \sum_{s=1}^{t-1} \langle \nabla f_s(x_s), x \rangle + \frac{1}{\eta} R(x) \right]$$

where  $R(x)$  is a strongly convex regularizer.

**Common choices:**

- $R(x) = \frac{1}{2} \|x\|^2$  (quadratic)
- $R(x) = \sum_i x_i \log x_i$  (entropy)

**Intuition:** Balance fitting past gradients against staying close to "simple" solutions.

## 2. Regret Bound

**Theorem:** FTRL with 1-strongly convex regularizer  $R$  achieves:

$$R_T \leq \frac{R(x^*)}{\eta} + \frac{\eta}{2} \sum_{t=1}^T \|\nabla f_t(x_t)\|^2$$

Optimizing  $\eta$  gives  $O(\sqrt{T})$  regret.

**Advantage:** More general than OGD, captures many algorithms as special cases.

### 3. Connection to MWU

**MWU as FTRL:** Multiplicative Weights is FTRL with:

- $\mathcal{K} = \text{simplex}$
- $R(x) = \sum_i x_i \log x_i$  (negative entropy)
- Losses  $\langle \ell_t, x \rangle$

**Result:** MWU update  $w_{t+1}(i) \propto w_t(i) \exp(-\eta \ell_{t,i})$  is exactly FTRL with entropy regularizer!

**Unifying principle:** FTRL explains why exponential weights work.

### 5. Strongly Convex Functions

#### 1. Definition

**$\alpha$ -strongly convex:** Function  $f$  satisfies for all  $x, y$ :

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle + \frac{\alpha}{2} \|y - x\|^2$$

**Interpretation:**  $f$  grows at least quadratically away from any point.

**Examples:**

- $\frac{1}{2} \|x\|^2$  is 1-strongly convex
- $\|x\|^2 + \lambda \|x\|_1$  is 2-strongly convex (for quadratic part)

#### 2. Why It Helps

**Key property:** Strong convexity ensures unique minimizer and fast convergence.

**In FTRL:** Strongly convex regularizer  $R$  ensures:

1. Unique solution  $x_t$  at each round
2. Stability across rounds (bounded movement)
3. Tighter regret bounds

**Trade-off:** Stronger  $\alpha \rightarrow$  tighter bounds but more restrictive.

### 6. Advanced Topics

#### 1. Adaptive Learning Rates

**Problem:** Optimal  $\eta$  depends on unknown  $T$ .

**Solutions:**

1. **Doubling trick:** Restart with doubled horizon

2. **AdaGrad:**  $\eta_t = \frac{\eta_0}{\sqrt{\sum_{s=1}^t \|\nabla f_s(x_s)\|^2}}$

3. **RMSprop/Adam:** Exponentially weighted running average

**Result:** Achieve  $O(\sqrt{T})$  regret without knowing  $T$ .

### 2. Bandit Feedback

**Full information:** Observe entire loss function  $f_t$

**Bandit:** Only observe  $f_t(x_t)$  (loss at played point)

**Challenge:** Must balance exploration and exploitation.

**Result:** Still possible to achieve  $O(\sqrt{T})$  regret, but with worse constants (additional  $\sqrt{d}$  factor).

### 3. Stochastic vs Adversarial

**Stochastic OCO:** Losses  $f_t$  drawn i.i.d. from distribution.

- Can achieve  $O(\frac{1}{\sqrt{T}})$  convergence in expectation
- Related to stochastic gradient descent (SGD)

**Adversarial OCO:** Worst-case sequence of losses.

- $O(\sqrt{T})$  regret is unavoidable
- But provides robustness guarantees

**Hybrid:** Best-of-both-worlds algorithms adapt to problem difficulty.

### 7. Key Pitfalls

**Convexity is crucial:** OGD and FTRL require convex losses. For non-convex problems, need different tools (e.g., multi-armed bandits, evolutionary strategies).

**Projection cost:** Computing  $\Pi_{\mathcal{K}}(y)$  can be expensive for complex  $\mathcal{K}$ . Choose simple constraint sets when possible.

**Strong convexity  $\neq$  convexity:** Strongly convex is strictly stronger. Not all convex functions are strongly convex (e.g., hinge loss).

**FTRL  $\neq$  OGD:** FTRL minimizes cumulative loss + regularizer. OGD does gradient step + projection. They're related but distinct.

**Learning rate matters:** Wrong  $\eta$  can cause regret to scale as  $O(T)$  instead of  $O(\sqrt{T})$ . Use theory to set  $\eta$  or adaptive methods.

**Diameter and gradient bounds:** Regret depends on  $D = \text{diam}(\mathcal{K})$  and  $G = \sup \|\nabla f_t\|$ . Large values  $\rightarrow$  large regret. Normalize inputs if possible.